VOSSA: Voiceprint Optimization for Streaming Speech Architectures

Mu-Ruei Tseng Waris Quamer Ghady Nasrallah Ricardo Gutierrez-Osuna

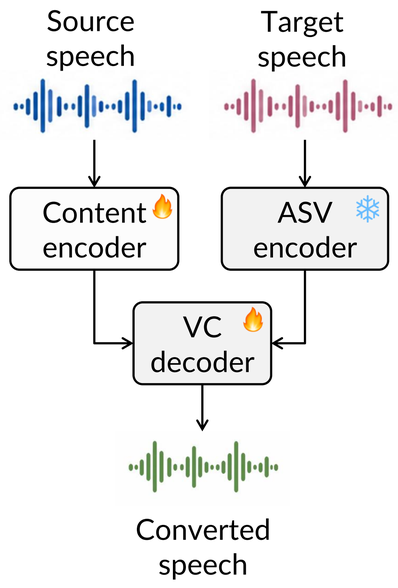
PSI Lab, Department of Computer Science and Engineering, Texas A&M University, USA

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Motivation

Most streaming VC models rely on automatic speaker verification (ASV) encoders to extract speaker embeddings.

- ASV embeddings are optimized for speaker discrimination, not speech generation.
- They are learned independently of the VC generator, which presents a potential task mismatch.



Can a speaker representation be learned jointly with the speech generation task?

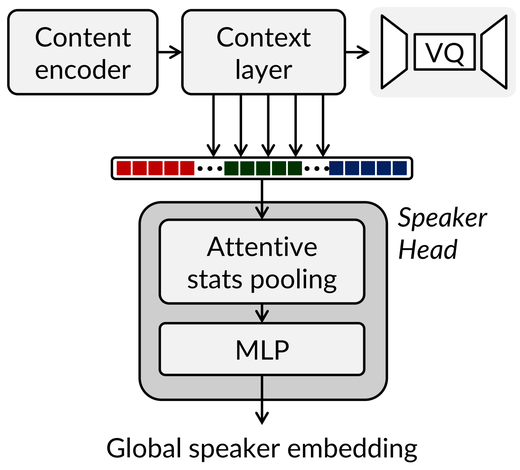
Proposed Method

VOSSA eliminates the external ASV speaker encoder by deriving speaker representations directly from intermediate features of the frozen streaming content encoder. The speaker representation module is trained jointly with the VC model.

1. Speaker representation from multi-level content features

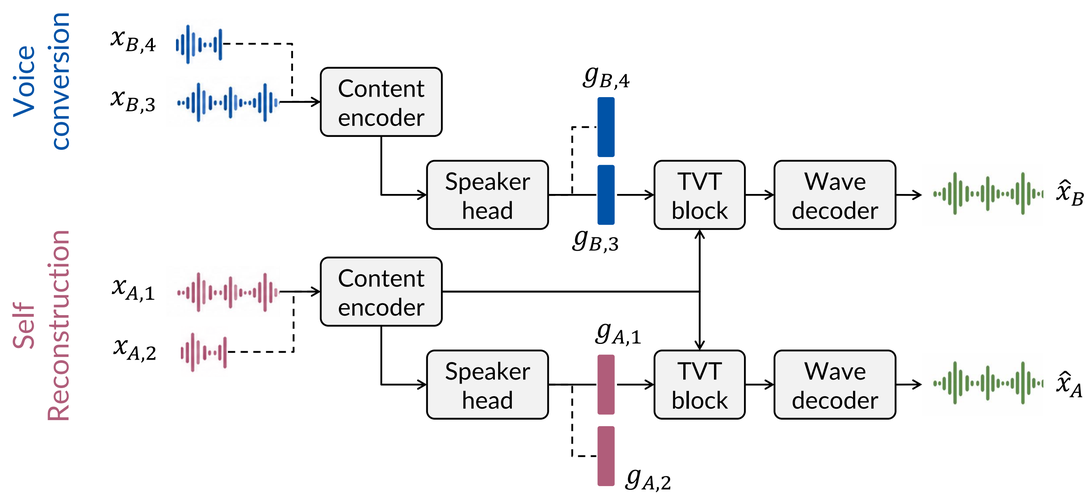
VOSSA probes multiple intermediate layers of a causal content encoder (TVTSyn)

$$\mathbf{H}_t = \mathrm{Concat}\left(\mathbf{h}_t^{\mathrm{CNN}}, \mathbf{h}_t^{(2)}, \mathbf{h}_t^{(4)}, \dots\right) \in \mathbb{R}^{Ld}, \quad g(x) = \mathrm{MLP}\left(\mathrm{ASP}(\mathbf{H})\right).$$



Combines complementary speaker cues from multiple encoder depths without requiring a separate speaker encoder.

2. Dual-path speaker-aligned training



For each speaker $S \in \{A, B\}$, we average embeddings from two utterances to form a stable speaker reference:

$$\bar{g}_S = \frac{g(x_{S,1}) + g(x_{S,2})}{2}.$$

Given source content $c_A = E_c(x_{A,1})$, VOSSA jointly optimizes two paths:

$$\underbrace{\hat{x}_A = D(c_A, \bar{g}_A)}_{\text{self-reconstruction}}, \qquad \underbrace{\hat{x}_B = D(c_A, \bar{g}_B)}_{\text{non-parallel conversion}}.$$

- **Self-reconstruction** provides direct acoustic supervision while preserving the source identity.
- **Non-parallel conversion** transfers the same content to the target identity without parallel source-target speech.

Training Objective

We design a joint objective that balances acoustic reconstruction with target-speaker identity preservation:

$$\mathcal{L}_{\text{total}} = \overbrace{\lambda_{\text{mel}}\mathcal{L}_{\text{mel}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{fm}}\mathcal{L}_{\text{fm}}}^{\text{Speech generation}} + \overbrace{\lambda_{\text{spk}}\mathcal{L}_{\text{spk}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}}}^{\text{Speaker alignment}}.$$

- Reconstruction losses: Mel, adversarial, and feature-matching losses supervise only the self-reconstruction path.
- Speaker consistency: $\mathcal{L}_{\text{spk}}$ preserves the intended target identity.

$$\mathcal{L}_{spk} = \cos(g(\hat{x}_A), \bar{g}_A) + \cos(g(\hat{x}_B), \bar{g}_B)$$

- Contrastive learning: Symmetric InfoNCE loss $\mathcal{L}_{\text{con}}$ structures the speaker embedding space.

$$\mathcal{L}_{con} = \mathrm{InfoNCE}_{sym}(g_{A,1}, g_{A,2}, g_{B,3}, g_{B,4})$$

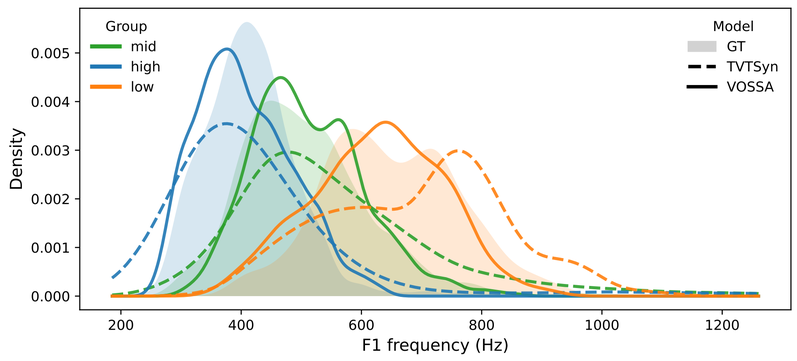
Results

1. Voice conversion and pitch

Model	NISQA↑	WER↓	$\mathbf{Sim}_{trg}^{syn}$↑	Pitch MAE↓	PCC↑
slt24	3.46	0.18	0.46	31.4	0.40
DarkStream	3.12	0.25	0.54	33.3	0.21
GenVC-s	3.04	0.20	–	30.8	0.33
TVTSyn	**3.52**	**0.17**	0.59	29.0	0.40
VOSSA	3.48	**0.17**	**0.86**	**27.8**	**0.43**

VOSSA maintains quality and intelligibility while improving target-speaker similarity and pitch dynamics.

2. Speaker-dependent acoustic realization



F1 distributions by vowel height for VoxCeleb speaker id00061. VOSSA more closely follows the ground-truth distributions.

	slt24	DarkStream	TVTSyn	VOSSA
High vowels ↓	42.3	28.9	29.2	**24.6**
Mid vowels ↓	44.5	31.9	32.9	**29.7**
Low vowels ↓	57.1	37.4	36.6	**31.0**

VOSSA achieves the lowest F1 Wasserstein distance across all vowel heights, indicating better speaker-dependent phonetic-acoustic realization.

3. Perceptual evaluation and streaming efficiency

Listener preference	TVTSyn	VOSSA
Speaker similarity	46%	**54%**
Intelligibility	44%	**56%**
Vibrancy	48%	**52%**

VOSSA is preferred across all perceptual criteria and achieves the highest subjective MOS among synthesized systems (3.79 ± 1.01).

132.4M parameters | 19% reduction | 73 ms latency | RTF 0.25

Conclusion

- Intermediate content features provide effective speaker conditioning without a standalone speaker encoder.
- VOSSA improves speaker transfer, pitch dynamics, and vowel-dependent acoustics while preserving quality and intelligibility.
- VOSSA reduces model size by 19% with no additional streaming overhead.

Takeaway: Speaker representations for synthesis can be learned jointly with the generator rather than imported from speaker recognition.